

Comparison of Apriori Algorithm for Frequent Itemset Mining with State-of-the-Art Algorithms and Optimization Strategies

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Abstract—to be written in the end.

I. INTRODUCTION

Frequent Itemset Mining (FIM) stands as a foundational pillar within the domain of data mining, tasked with uncovering sets of items that consistently co-occur across massive transactional databases. The extraction of these patterns is critical for generating actionable association rules, which drive decision-making processes across a multitude of industries. For instance, in retail analytics, FIM powers market basket analysis to uncover product bundling opportunities. In healthcare informatics, it facilitates the discovery of co-occurring symptoms or medication patterns within clinical records. Furthermore, FIM is instrumental in financial sectors for fraud detection by isolating suspicious, recurring transactional behaviors.

Despite its immense utility, FIM remains computationally demanding. The classical Apriori algorithm [1], while conceptually elegant, relies on a breadth-first search and a generate-and-test paradigm that introduces severe performance bottlenecks. Primarily, Apriori suffers from exorbitant I/O costs, as it necessitates a full database scan to compute support counts for every newly generated level of candidates. Furthermore, in dense datasets or environments with low minimum support thresholds, the algorithm triggers an exponential explosion in candidate generation, rapidly exhausting memory and computational resources.

To address these challenges, this study rigorously evaluates the classical Apriori algorithm against a contemporary, state-of-the-art frequent itemset mining framework published post-2022. By juxtaposing the traditional horizontal data format against modern vertical TID-list representations, we aim to dissect the evolutionary improvements in algorithm design.

The core contributions of this paper are threefold. First, we establish a robust empirical baseline by evaluating Apriori against a recent vertical-format algorithm on dense benchmark datasets. Second, we propose and implement two distinct optimization strategies designed to directly mitigate the baseline's support counting bottlenecks. Finally, we provide a comprehensive comparative analysis of execution time, memory

consumption, and scalability to highlight the practical trade-offs of modern algorithmic enhancements.

II. LITERATURE REVIEW

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III. ALGORITHMS: DESCRIPTION AND ANALYSIS

waiting for colleague to submit the analysis.

IV. PROPOSED OPTIMIZATION STRATEGIES

waiting for colleague to submit the explanation.

V. EXPERIMENTAL SETUP AND RESULTS

waiting for colleagues to submit the figures findings.

VI. DISCUSSION

yet to be written.

VII. CONCLUSION

yet to be written.

REFERENCES

- [1] R. Agrawal and R. Srikant, "Fast algorithms for mining association rules," in *Proc. 20th Int. Conf. Very Large Data Bases (VLDB)*, 1994, pp. 487–499.